

Organizing a Java Project

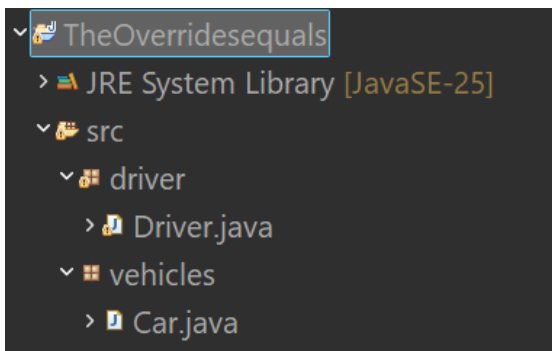
Course Document

How Eclipse's project view corresponds to your folders on disk

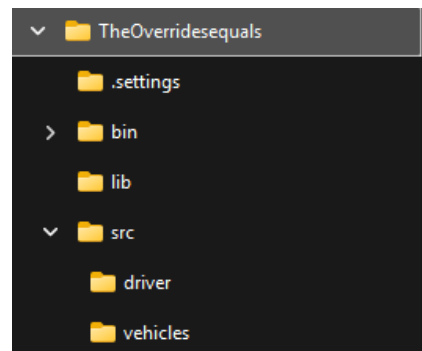
In this course, unless an assignment says otherwise, you may use any IDE or editor to write your Java code, such as Eclipse, VS Code, Neovim, Emacs, or even a plain text editor. Regardless of the editor, every project must be organized the same way Eclipse organizes a project on disk. The images below show the same project two ways: how Eclipse displays it in the Package Explorer, and how it looks as plain folders in the file system.

Recommendation: Until you are comfortable with project organization and source control with GitHub, I recommend using Eclipse. Projects are easier to organize and commit correctly in Eclipse than in other tools.

1. How the Views Correspond



How Eclipse shows the project



The same project as plain folders

The example above shows a project named “TheOverrideequal” with source folder containing two packages, driver and vehicles. In those packages, there are .java files. This is the basic layout, but the project, package, and file names will vary depending on the assignment.

Table 1. How each item in Eclipse's Package Explorer maps to a real folder or file.

Eclipse Package Explorer	File System	What It Is
Project (e.g. lastlabw01)	lastlabw01/	The top-level folder. Name it exactly as the assignment instructs for a project name.
src	lastlabw01/src/	Holds only package folders. No .java files belong directly inside it.
package driver	lastlabw01/src/driver/	A package is just a subfolder inside src. Package names are lowercase by convention.
Driver.java	lastlabw01/src/driver/Driver.java	A source file, placed inside its package's folder.
package vehicles	lastlabw01/src/vehicles/	Additional packages are additional subfolders inside src.

Eclipse Package Explorer	File System	What It Is
and so on ...	and so on ...	Repeat this pattern for every package and source file in the project.

2. Folder and File Rules

Package names: A package name that includes dots, such as `edu.nsu.cis3300`, becomes nested folders on disk: `edu/nsu/cis3300`.

bin folder: Not required in your submission. I compile your source code myself. It is fine to include it, but it is unnecessary.

.settings folder: Not required either. It is fine if you include it.

lib folder: Include one only if your project uses external libraries. Whether this applies depends on the assignment.

Version control: Committing an Eclipse project with Git usually captures everything needed. If you use a different IDE, double-check what it is tracking before you commit. See the *Setting Up GitHub and Your Development Environments* guide for staging and committing steps in both Eclipse and VS Code.

3. Using a Non-Eclipse IDE or Editor

If you are not using Eclipse, you do not need to reproduce Eclipse's Package Explorer icons or metadata files. Your file manager or IDE's file tree only needs to match the right-hand image above: a project folder containing a `src` folder, with one subfolder per package, and your `.java` files inside those package folders.